

Research Sub-Group: 3-Person Delegation & Execution Plan

Adaptive Attention Computation • Theory, Kernel Engineering & Model Integration

Target Horizon: 8.5 – 9.5 Hours	Sub-Group Size: 3 Members (Theory, Kernel, Integration)	Partner Team: Testing & Benchmarking Group
Primary Hardware: Kaggle Dual T4 (16GB VRAM)	Baseline Model: Qwen/Qwen2.5-0.5B (FP16)	Core Output: adaptive_kernel.py, patch_qwen.py, Deck

1. Purpose & Delegation Strategy

While the external Testing Team focuses on empirical compute profiling, Passkey retrieval suites, and SOTA DeltaNet baselines, the Research & Modeling team is responsible for theoretical formulation, PyTorch kernel implementation, and pretrained LLM patching. To ensure rapid, parallel execution with zero workflow blockages, the research responsibilities are decoupled into **three specialized roles: Member 1 (Theory & Mathematical Bounds), Member 2 (Kernel & Systems Engineering), and Member 3 (Model Integration & Deck Synthesis).**

2. 3-Person Role Profiles & Delegatable Workstreams

Dimension	Member 1: Theory & Bounds	Member 2: Kernel Engineer	Member 3: Model Integration
Core Focus	Mathematical error bounds, input entropy metrics, and analytical proofs.	Production PyTorch kernel implementation, stability, and unit tests.	LLM attention monkey-patching, prefill profiling, and deck assembly.
Primary Deliverables	- difficulty_metric.py - Math proof notes (LaTeX) - Theory slide narratives	- adaptive_kernel.py - test_adaptive_kernel.py - Micro-benchmark script	- patch_qwen.py - Prefill latency benchmark CSV - Master 7-Slide Deck & PDF
Key Dependencies	Provides error bounds & rank criteria to Member 2.	Ingests bounds from M1; delivers stable kernel to M3 & Testing Team.	Ingests kernel from M2; integrates with Qwen2.5 and testing results.
Success Metric	Exact bound certificate: $\epsilon_r \leq \delta Z_i / [N(2 V_{\max} + \delta)]$	Agreement $K \approx L$ to $< 10^{-15}$; zero causal leakage ($\Delta = 0.00$).	Patched Qwen2.5 forward pass succeeds with < 1.5 GB VRAM footprint.

3. Technical Specifications & Interface Contracts

A. Kernel Function Signature Contract (Member 2 → Member 3 & Testing Team):

```
def adaptive_rank_attention(Q, K, V, delta=0.05, r_candidates=(64, 256, 1024), is_causal=False) -> torch.Tensor
```

- Inputs: Q, K: (B, H, N, d_k); V: (B, H, N, d_v) | Output: (B, H, N, d_v)
- Numerical Safeguards: FP32 accumulator enforcement for FP16/BF16 inputs; epsilon clamp: (den + 1e-6).

B. Error Certificate & Difficulty Estimator Contract (Member 1 → Member 2):

```
def estimate_required_rank(Q, K, delta=0.05, r_candidates=(64, 256, 1024)) -> int
```

- Computes query-key logit entropy: $H = -\sum(p * \log(p))$ and temperature scale.
- Allocates rank: $r = 64$ if $H > 0.85$ (diffuse); $r = 256$ if $0.4 < H \leq 0.85$; $r = 1024$ if $H \leq 0.4$ (sharp needle).

4. Master Hour-by-Hour Schedule & Handoff Checkpoints

The sprint runs concurrently with the Testing Team's benchmarking schedule. Strict handoff gates ensure that Member 2's kernel is tested before Member 3 deploys it into Qwen2.5.

Window	Member 1: Theory Lead	Member 2: Kernel Engineer	Member 3: Integration Lead	Handoff & Sync Gate
0.0 – 1.0 h <i>Phase 1</i>	- Formulate adaptive error bound equation - Lock rank selection rule - Outline Slide 2 & 5	- Publish MockKernel stub - Freeze function I/O signature - Lock CSV schema contract	- Setup Qwen2.5 environment - Validate baseline SDPA inference - Outline master deck slides	GATE 1 (Hr 1.0) Interface Lock Testing Team unblocked with MockKernel.
1.0 – 3.5 h <i>Phase 2</i>	- Implement difficulty_metric.py - Verify 3-token analytical check ([4.8, 5.2]) - Prove uniform bounds	- Implement ReLU+1 (r=64) and FAVOR+ (r=256, 1024) - Enforce FP32 accumulators - Write test harness	- Author patch_qwen.py monkey-patch wrapper - Verify replacement hook on dummy Linear Attention	GATE 2 (Hr 3.5) Alpha Kernel Member 2 delivers alpha kernel to Member 3.
3.5 – 6.0 h <i>Phase 3</i>	- Draft Slide 1 (Dilemma), Slide 2 (Hypothesis), and Slide 5 (Gap) - Write LaTeX math appendix	- Add causal prefix-sums ($S_t = S_{t-1} + \phi(k_t)v_t^T$) - Execute full unit test suite - Micro-benchmark FLOPs	- Drop alpha kernel into Qwen2.5-0.5B - Verify forward pass & memory - Setup prefill benchmark harness	GATE 3 (Hr 6.0) Beta Kernel Patched Qwen2.5 running. Mid-sprint sync with Testing Team.
6.0 – 7.5 h <i>Phase 4</i>	- Review and sign-off on kernel mathematical correctness - Draft Slide 6 & 7 narratives	- Package production adaptive_kernel.py - Optimize memory and CUDA event synchronization	- Benchmark prefill latency across N = 1K to 8K on Kaggle T4 - Ingest initial test plots	GATE 4 (Hr 7.5) CODE FREEZE Production kernel handed to Testing Team.
7.5 – 8.5 h <i>Phase 5</i>	Author theoretical analysis for executive report	Code audit & repository clean-up	Ingest final Testing CSVs & figures into slide deck	GATE 5 (Hr 8.5) Deck Audit All metrics cross-checked.
8.5 – 9.5 h <i>Phase 6</i>	Rehearse theory and math defense	Package code archive & reproduction scripts	Lead presentation dry-run and executive sign-off	FINAL DELIVERY Hr 9.5 Project complete.

INTERNAL HANDOFF LIFECYCLE:

M1 (Math Formula) → M2 (adaptive_kernel.py) → M3 (patch_qwen.py) → Testing Team (Benchmarks) → M3 (Master Deck Assembly)

5. Internal RACI Responsibility Matrix

Deliverable / Workstream	M1 (Theory)	M2 (Kernel)	M3 (Integration)	Sign-Off Lead
Analytical Error Bound Derivations & LaTeX Spec	Responsible (R)	Consulted (C)	Informed (I)	Member 1
Difficulty / Entropy Estimator (difficulty_metric.py)	Responsible (R)	Consulted (C)	Informed (I)	Member 1
Production Kernel Implementation (adaptive_kernel.py)	Consulted (C)	Responsible (R)	Consulted (C)	Member 2
FP32 Stability Safeguards & Denominator Epsilon Clamping	Consulted (C)	Responsible (R)	Informed (I)	Member 2
Automated Unit Test Suite (test_adaptive_kernel.py)	Consulted (C)	Responsible (R)	Informed (I)	Member 2
Qwen2Attention Forward Monkey-Patch (patch_qwen.py)	Informed (I)	Consulted (C)	Responsible (R)	Member 3
Prefill Latency & Memory Profiling on Qwen2.5-0.5B	Informed (I)	Consulted (C)	Responsible (R)	Member 3
Master 7-Slide Deck Assembly & Executive Narrative	Consulted (C)	Informed (I)	Responsible (R)	Member 3
Interface Contract Sign-Off with External Testing Team	Consulted (C)	Responsible (R)	Responsible (R)	M2 & M3

6. Operational Guardrails & Systems Dependencies

Constraint / Area	Engineering Specification & Implementation Rule	Owner
Accumulator Precision	When inputs are FP16 or BF16, force all intermediate kernel accumulators (KV state and denominator sum) to torch.float32 to prevent overflow across long sequences ($N \geq 8,192$).	Member 2
Model Memory Cap	Patching must be strictly tested on Qwen/Qwen2.5-0.5B (~1.0 GB VRAM in FP16), leaving > 12 GB VRAM free on the 16GB Tesla T4 for sequence prefill tensors.	Member 3
Testing Interface	All kernel implementations must conform to the frozen function signature and accept kwargs delta=0.05 and is_causal=False without exceptions.	Member 2

7. Research Sub-Group Deliverables Checklist

[]	1. Mathematical Specification: difficulty_metric.py with unit-tested entropy estimator and formal LaTeX error bound proofs.
[]	2. Production Adaptive Kernel: adaptive_kernel.py supporting dynamic ranks $r \in \{64, 256, 1024\}$ and causal prefix-sums.
[]	3. Kernel Verification Suite: test_adaptive_kernel.py passing all analytical checks, numerical agreement ($K \approx L$), and causal tests.
[]	4. Pretrained Model Patch: patch_qwen.py monkey-patching Qwen2.5-0.5B and profiling prefill latency up to 8K.
[]	5. Master Presentation & Summary: Consolidated 7-slide deck integrating empirical figures and final executive technical report.